# Snowflake Project – Data Engineering & SQL Solution

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| Date | 01-11-2026 |
| Team ID | EN25CS3011097 |
| Project Name | Global Olympic Sports Performance Analysis |

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission demonstrates that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis.

## Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

## Solution Summary

| Field | Details |
|---|---|
| Project / Repository Link | To be added by the team (GitHub link) |
| Data Source(s) Link | athelete_events.csv – Olympic dataset 1992–2026 (8,988 rows × 56 columns; record types: athletes 8,000, medal_results 910, alltime_country_rankings 60, games_summary 18), uploaded to stage OLYMPIC_STAGE |
| Database | OLYMPIC_DB (schema SPORTS) |
| Warehouse | OLYMPIC_WH – size XSMALL, AUTO_SUSPEND = 60 seconds, AUTO_RESUME = TRUE |
| Tables / Views Created | Table OLYMPIC_EVENTS (56 columns); file format CSV_FF; stage OLYMPIC_STAGE; staging views STG_ATHLETES, STG_MEDAL_RESULTS, STG_COUNTRY_RANKINGS, STG_GAMES_SUMMARY; cleaned view CLN_OLYMPIC_EVENTS; mart views MART_COUNTRY_MEDALS, MART_SPORT_PARTICIPATION |
| Key Transformations / Calculations | Filter on dataset_type for every record type; trim and standardize text; derived Total Medals Calculated = gold + silver + bronze; SUM, COUNT(DISTINCT athlete_id), GROUP BY, ORDER BY and LIMIT for Top 10 rankings |
| KPIs created | Total medals by country; gold / silver / bronze split; total medal points; athletes per sport; events per athlete; Key Figures (60 countries, 50 sports, 8,000 athletes, 18 Games); 7 Tableau worksheets, Dashboard 1 and Story 1 |
| Key Insights | United States (735) and China (696) lead in total medals; gold race is close (250 vs 230); medal points US 1,443 and China 1,386; participation is evenly spread across 50 sports (Surfing 201 to Luge 100); Europe leads by region with 2,822 of 6,025 medals (46.8%) |

## Snowflake Quality Checklist

| S.No | Criteria | Status (Yes / No) |
|---|---|---|
| 1 | Required data is successfully loaded and accessible in Snowflake | Yes |
| 2 | Data is cleaned, transformed, and prepared appropriately | Yes |
| 3 | Required tables, views, and data relationships are correctly implemented | Yes |
| 4 | Required SQL logic, calculations, and KPIs are correctly implemented | Yes |
| 5 | Data and solution outputs are validated against the project requirements | Yes |
| 6 | Solution is properly organized, documented, and functional | Yes |

**Validation evidence (Snowflake test cases)**

| Test ID | Scenario | Expected result | Status |
|---|---|---|---|
| SF-01 | COPY INTO load | 8,988 rows loaded without errors | Pass |
| SF-02 | Count by dataset_type | 8,000 / 910 / 60 / 18 | Pass |
| SF-03 | Top 10 countries by medals | United States 735, China 696, Russia 429 | Pass |
| SF-04 | Athletes per sport | Surfing 201, Canoe/Kayak 199, Handball 193 | Pass |
| SF-05 | Medal points ranking | United States 1,443, China 1,386 | Pass |

**Additional Notes / Comments:**

- Blank values are structural: each dataset_type fills only its own columns, so every query filters on dataset_type.
- Query execution times are recorded from Snowflake Query History; the GitHub link will be added by the team.

## SQL Scripts

### Step 1: Create warehouse, database and schema

```
CREATE WAREHOUSE OLYMPIC_WH WITH WAREHOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE;
CREATE DATABASE OLYMPIC_DB;
CREATE SCHEMA OLYMPIC_DB.SPORTS;
USE WAREHOUSE OLYMPIC_WH;
USE SCHEMA OLYMPIC_DB.SPORTS;
```

### Step 2: File format and stage

```
CREATE OR REPLACE FILE FORMAT CSV_FF
  TYPE = CSV FIELD_DELIMITER = ',' SKIP_HEADER = 1
  FIELD_OPTIONALLY_ENCLOSED_BY = '"' NULL_IF = ('', 'NULL');
CREATE OR REPLACE STAGE OLYMPIC_STAGE FILE_FORMAT = CSV_FF;
PUT file://athelete_events.csv @OLYMPIC_STAGE;
```

### Step 3: Create the table (56 columns)

```
CREATE OR REPLACE TABLE OLYMPIC_DB.SPORTS.OLYMPIC_EVENTS (
    DATASET_TYPE VARCHAR, YEAR FLOAT, HOST_CITY VARCHAR, HOST_COUNTRY VARCHAR,
    SEASON VARCHAR, COUNTRY VARCHAR, NOC_CODE VARCHAR, REGION VARCHAR, GDP_TIER VARCHAR,
    POPULATION_MILLION FLOAT, SPORTS_INVESTMENT VARCHAR, ATHLETES_SENT FLOAT, GOLD FLOAT,
    SILVER FLOAT, BRONZE FLOAT, TOTAL_MEDALS FLOAT, MEDAL_POINTS FLOAT,
    MEDAL_EFFICIENCY_PCT FLOAT, TOP_SPORT VARCHAR, HOST_NATION FLOAT, TOTAL_ATHLETES FLOAT,
    PARTICIPATING_NATIONS FLOAT, TOTAL_MEDALS_AWARDED FLOAT, GOLD_MEDALS_AWARDED FLOAT,
    NATIONS_WON_MEDAL FLOAT, TOP_COUNTRY VARCHAR, ATHLETE_ID VARCHAR, NAME VARCHAR,
    GENDER VARCHAR, AGE_AT_PEAK FLOAT, SPORT VARCHAR, SPORT_CATEGORY VARCHAR,
    FIRST_GAMES_YEAR FLOAT, LAST_GAMES_YEAR FLOAT, GAMES_PARTICIPATED FLOAT,
    GOLD_MEDALS FLOAT, SILVER_MEDALS FLOAT, BRONZE_MEDALS FLOAT, WORLD_RECORD_HOLDER FLOAT,
    PERSONAL_BEST_RANK_WORLD FLOAT, HEIGHT_CM FLOAT, WEIGHT_KG FLOAT, BMI FLOAT,
    TRAINING_HOURS_PER_WEEK FLOAT, PROFESSIONAL_ATHLETE FLOAT, DOPING_VIOLATION FLOAT,
    MOST_RECENT_MEDAL_2020PLUS FLOAT, GAMES_ENTERED FLOAT, TOTAL_GOLD FLOAT,
    TOTAL_SILVER FLOAT, TOTAL_BRONZE FLOAT, TOTAL_MEDAL_POINTS FLOAT,
    AVG_EFFICIENCY_PCT FLOAT, SUMMER_MEDALS FLOAT, WINTER_MEDALS FLOAT, OVERALL_RANK FLOAT
);
```

### Step 4: Load and verify

```
COPY INTO OLYMPIC_EVENTS
FROM @OLYMPIC_STAGE/athelete_events.csv.gz
FILE_FORMAT = (FORMAT_NAME = CSV_FF)
ON_ERROR = 'ABORT_STATEMENT';

SELECT dataset_type, COUNT(*) FROM olympic_events GROUP BY dataset_type ORDER BY 2 DESC;
```

### Step 5: Data preparation views

```
CREATE OR REPLACE VIEW STG_ATHLETES AS SELECT * FROM OLYMPIC_EVENTS WHERE dataset_type='athletes';
CREATE OR REPLACE VIEW STG_MEDAL_RESULTS AS SELECT * FROM OLYMPIC_EVENTS WHERE dataset_type='medal_results';
CREATE OR REPLACE VIEW STG_COUNTRY_RANKINGS AS
  SELECT * FROM OLYMPIC_EVENTS WHERE dataset_type='alltime_country_rankings';
CREATE OR REPLACE VIEW STG_GAMES_SUMMARY AS SELECT * FROM OLYMPIC_EVENTS WHERE dataset_type='games_summary';
-- CLN_OLYMPIC_EVENTS, MART_COUNTRY_MEDALS and MART_SPORT_PARTICIPATION: see the Data Preparation report
```

### Step 6: Analysis queries (one for each Tableau view)

```
-- Query 1: Top 10 countries by total medals
SELECT country, SUM(total_medals) AS total_medals FROM olympic_events
WHERE dataset_type='medal_results' GROUP BY country ORDER BY total_medals DESC LIMIT 10;

-- Query 2: Gold, silver and bronze by country
SELECT country, SUM(gold) AS gold, SUM(silver) AS silver, SUM(bronze) AS bronze FROM olympic_events
WHERE dataset_type='medal_results' GROUP BY country ORDER BY SUM(total_medals) DESC LIMIT 10;

-- Query 3: Top 10 sports by number of athletes
SELECT sport, COUNT(DISTINCT athlete_id) AS athletes FROM olympic_events
WHERE dataset_type='athletes' GROUP BY sport ORDER BY athletes DESC LIMIT 10;

-- Query 4: Top 10 athletes by number of events
SELECT name, COUNT(DISTINCT athlete_id) AS events FROM olympic_events
WHERE dataset_type='athletes' GROUP BY name ORDER BY events DESC, name LIMIT 10;

-- Query 5: Top 10 countries by medal points
SELECT country, total_medal_points FROM olympic_events
WHERE dataset_type='alltime_country_rankings' ORDER BY total_medal_points DESC LIMIT 10;
```